database should be used for evaluation purposes only.
4. Configure the database.
5. Extract the downloaded SonarQube archive to a directory of your choice.
6. Start the SonarQube server. Navigate to the directory where you extracted the archive and run the following command.

Linux:

```
sonarqube-8.9.10.61524/bin/linux-x86-64$ sh sonar.sh start
osfy@ubuntu:~/Desktop/OSFY/March2023/#tools/
sonarqube-8.9.10.61524/bin/linux-x86-64$ sh sonar.sh start
Starting SonarQube...
Started SonarQube.
```

Windows:

```
sonarqube-8.9.10.61524/bin/windows-x86-64/StartSonar.bat
```

7. Open a web browser and navigate to *http://localhost:9000*. You should see the SonarQube login page.
8. Log in to SonarQube. The default login credentials are *admin* for the user name and for the password. You can change the credentials after logging in for the first time.

That's it! You have now installed and started SonarQube.

Quality gates and quality profiles

In SonarQube, quality gates and quality profiles are used to define the acceptance criteria for code quality.

Quality gates: This is a set of predefined conditions that must be met before code can be considered good to go. For example, a quality gate may require that the code has a certain level of code coverage (such as 80 per cent), or that all vulnerabilities have to be addressed. If a quality gate is not met, the CI/CD pipeline must fail, and the development team must address the issues before it can go ahead.

Quality profiles: This is a collection of rules that define the expected quality standards.

Quality profiles can be customised to meet the specific needs of a project or organisation, such as rules can be activated or deactivated or configured as false positive.

By using quality gates and quality profiles, SonarQube makes it easy for teams to enforce best practices and maintain a high level of code quality. The results of the code analysis are automatically evaluated against the quality gates and profiles, ensuring that the code meets the standards set by the quality team and the CI/CD pipeline considers the outcome for next stage execution.

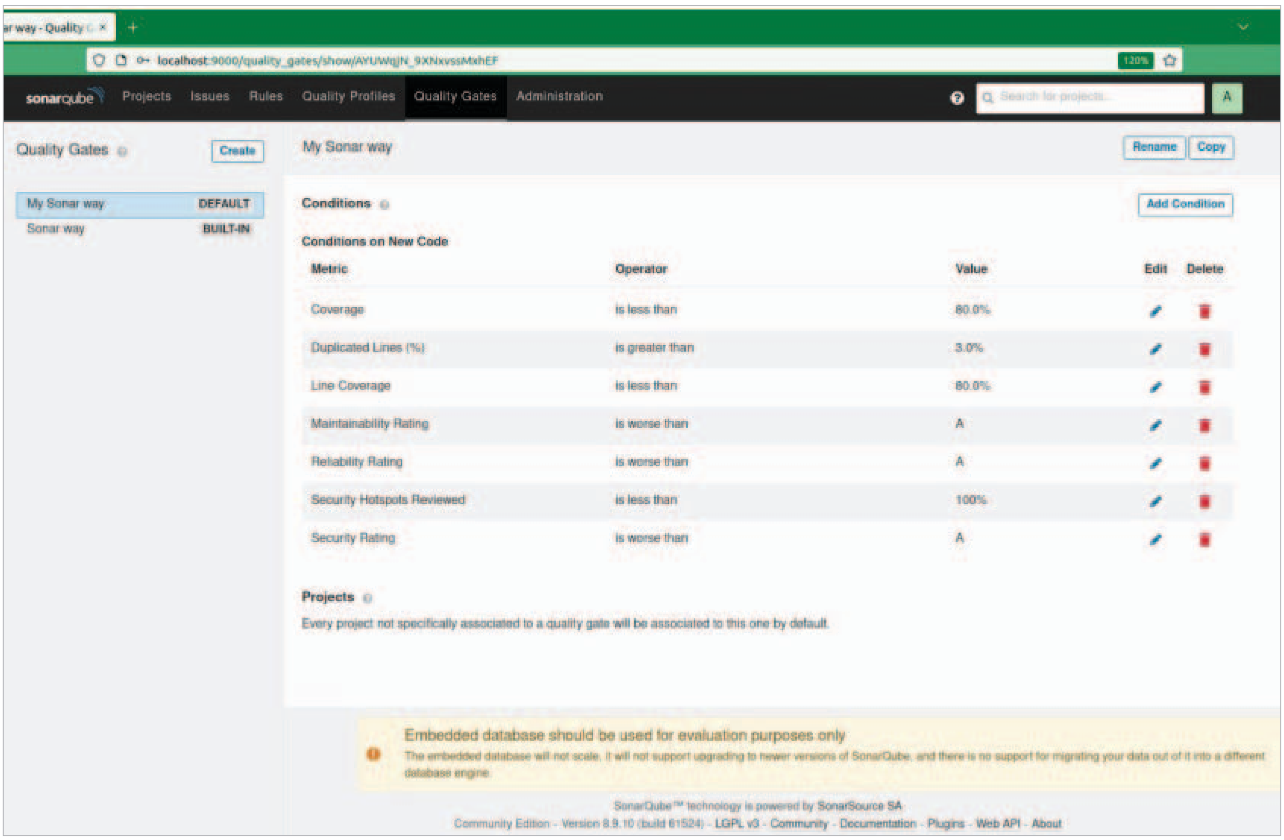

Figure 3: SonarQube quality gate